Tamal Batabyal

AI researcher | Machine Learning, Computational Neuroscience

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PROFILE

AI Researcher with experience developing computer vision, machine learning and deep learning models, graph-theoretic frameworks for large-scale neural and biomedical datasets. Proven record of translating fundamental research into deployable computational tools, publishing in leading IEEE and neuroscience journals, and collaborating across multidisciplinary teams.

TECHNICAL SKILLS

Programming : Python, MATLAB
Libraries : PyTorch, Scikit-learn, Pandas
AI/ML: Signal Processing, Image/Video Processing, Machine Learning, Deep Learning, CNN, Attention networks
Data modality : EEG, Local field potential (LFP), spikes, MRI, Ca-sensor data, Kinect, InSAR images (ArcGIS)

AI | ML RESEARCH EXPERIENCE

Massachusetts Institute of Technology — Aug. 2022 – Present
Postdoctoral Associate (Neuroscience), Advised by Earl K. Miller — Cambridge, MA

- Developed computational models for large-scale neuronal activity using dynamical systems in latent space for predictive coding. Found population-level multi-spectral signatures and topological separation of local and global oddball information in different brain regions.
- Showed correspondence between latent representation and the emergent traveling waves and linked them to cognitive functions and behavioral performance.
- Conducted training, invasive surgery and electrophysiological recordings in non-human primates.

University of Virginia — Aug. 2019 – June. 2022
Postdoctoral Associate (Neurology), Advised by Dr. Jaideep Kapur — Charlottesville, VA

- Developed CNN-attention net based model (Neurocounter) to detect and localize neurons from microscopic images (94% F-1 score, at least 3% reduction in false-positive detection compared to SOTA). Neurocounter was trained with a manually-curated, large-scale dataset with neuronal cellbodies and background artifacts. Observed self-learning behavior when trained with incompletely-annotated dataset.
- Configured the **closed-loop** optogenetics pipeline in the lab and developed deep-learning models for real-time seizure-**onset** detection (sequence modeling). Utilizing this pipeline and the model, conducted deep-brain stimulation using chemogenetics and optogenetics tools to modulate convulsive seizures.

University of Virginia — Aug. 2014 – May, 2019
Graduate Research Assistant, Advised by Scott T. Acton — Charlottesville, VA

- Built graph-theoretic ML models (graph-matching, probabilistic modeling) for automated catgegorization of neurons based on their morphology. Achieved at least 6% improvement in classification relative to SOTA.
- Worked in a team with Virginia DOT, USA and TRE Canada on a large-scale 2D+t point cloud of 3.6 million persistent scatterers per area of interest (InSAR radar images) to detect and characterize geohazards (sinkhole, pavement-rutting etc.). Developed a multiscale graph-theoretic framework that improved geohazard detection at least 10 times faster than competing approaches.
- (**Decorrelation**) Developed PreCoG, a graph-theoretic generalized decorrelator (split preconditioner) for streaming data. Achieved faster convergence (run-time improvement: at least 1.5 times for AR, 7 times for Linear systems) of transform-domain LMS filter weights than the SOTA for input data generative models (such as, autoregressive (AR) models) and linear systems of equations. When coupled with Hebbian learning, PreCoG showed significantly faster convergence for LMS filter weights.

TATA Consultancy Services Innovation labs May. 2012 – Aug, 2014
Intern Kolkata, India

- Worked in a team on developing a model mapping high-dimensional covariance to features on a low-dimensional linear manifold for real-time human activity classification using 3D Kinect skeleton data. Our approach outperformed competing approaches by at least 2%. (patented and published)

PUBLICATIONS (*selected*)

14 peer-reviewed journals, 12 peer-reviewed conferences

An efficient convolutional neural network for coronary heart disease prediction. Expert Systems with Applications, 2020, Key aspects: *AI in healthcare, CNN, Lasso*

Distinct roles of rodent thalamus and corpus callosum in seizure generalization. Annals of Neurology, 2022, Key aspects: *Clinical model, EEG*

Efficient learning of transform-domain LMS filter using graph Laplacian. IEEE transactions on neural networks and learning systems, 2022, Key aspects: *Graph theory, LMS filter, Hebb-LMS learning, Matrix perturbation*

Statespace trajectories and traveling waves following distraction. Journal of Cognitive Neuroscience, 2026, Key aspects: *Computational neuroscience, Spikes, Traveling waves, Subspace coding*

Neuropath2path: classification and elastic morphing between neuronal arbors using path-wise similarity. Neuroinformatics, 2020, Key aspects: *Computational neuroscience, Graph-matching, Self-similar system*

EDUCATION

University of Virginia Aug. 2014 – May 2019
Ph.D. in Electrical Engineering Charlottesville, VA

Indian Statistical Institute June 2012 – May 2014
M.Tech in Computer Science Kolkata, India

Indian School of Mines July 2006 – May 2010
B.Tech in Electrical Engineering Dhanbad, India

REVIEWER (*selected*)

IEEE Transaction on Image Processing, IEEE Transaction on Medical Imaging, IEEE Transaction on Pattern Analysis and Machine Intelligence, IEEE Signal Processing Letters, Journal of Neuroscience Methods, Expert systems with applications, Nature scientific reports, Neuroinformatics (Springer), IEEE Journal of Biomedical and Health Informatics, IEEE International Conference on Image Processing, IEEE International Symposium on Biomedical Imaging.